



VPC Traffic Flow and Security



Demayne Govender

sg-0a0d2e1421201e9d8 - Nextwork-security-group-public1

Security group (sg-0a0d2e1421201e9d8 | Nextwork-security-group-public1) was created successfully

Details

sg-0a0d2e1421201e9d8 - Nextwork-security-group-public1

Actions

Details

Security group name	Security group ID	Description	VPC ID
Nextwork-security-group-public1	sg-0a0d2e1421201e9d8	A Security Group for the Public1 VPC	vpc-0ce89e2934bb6614c
Owner	Inbound rules count	Outbound rules count	
479887546653	1 Permission entry	1 Permission entry	

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC(Virtual Private Cloud) is an isolated virtual network within AWS where we can launch and organise our cloud resources. VPC's are useful because they give us full control over our network environment including subnets, route tables, internet gateways and security settings while keeping our resources private and secure within the public cloud.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to build a secure network environment from the ground up. I configured a custom route table to direct internet-bound traffic through an internet gateway, making my subnet public. I then secured the network using two layers of protection. A security group acting as a virtual firewall at the resource level, and a custom network ACL applying broad traffic rules at the subnet level.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was how many moving parts work together to control traffic in a VPC. I assumed that a security group alone would be enough, but learned that network ACLs add a separate layer at the subnet level and that they are stateless while security groups are stateful. This showed me that cloud security is far more layered than I initially thought.



Demayne Govender

NextWork Student

nextwork.org

This project took me...

This project took me 2 hours to complete.



Route tables

Route tables act like a GPS for traffic within our VPC. It contains a set of rules called routes that determine where outbound traffic from a subnet gets directed. Each route has two parts: a destination(where the traffic is headed) and a target(what it should be sent through). For example, traffic destined for the internet gets routed through an internet gateway, while internal VPC traffic stays local. Every subnet must be associated with a route table, and it's the route table that determines whether a subnet is public or private. A subnet with a route to an internet gateway is public and one without it is private.

Route tables are needed to make a subnet public because simply attaching an internet gateway to a VPC is not enough because traffic also needs a path to reach it. That path is defined in the route table as a route from 0.0.0.0/0 to the gateway, which tells the subnet to send all internet-bound traffic through it. Without that route, traffic has no direction and never reaches the gateway, keeping the subnet private.

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/24	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute
0.0.0.0/0	igw-0230bd3e47bf5451Q			



Route destination and target

The destination is the IP address range the traffic is trying to reach. The target is the specific gateway or resource the traffic gets sent through to get there. For example, a destination of 0.0.0.0/0 with a target of igw-xxx means all internet-bound traffic is routed through the internet gateway. A destination matching the VPC's IP range with a target of local means the traffic stays within the VPC and communicates internally between resources.

The route in my route table that directed internet-bound traffic to my internet gateway had a destination of 0.0.0.0/0 and a target of igw-0230bd3e47bf54510. This means that all internet-bound traffic from the subnet is directed through my internet gateway.

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/24	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute

Buttons: Add route, Remove, Cancel, Preview, Save changes



Security groups

Security groups are virtual firewalls associated with individual resources inside a VPC, like EC2 instances. They control exactly what traffic is allowed in and out by filtering based on IP address, protocol, and port number. Unlike IAM policies, security groups only have 'allow' rules. Anything not explicitly allowed is automatically blocked. They are also stateful, meaning if inbound traffic is allowed, the response is automatically permitted back out without needing a separate outbound rule.

Inbound vs Outbound rules

Inbound rules are rules that control the data that can enter the resources in my security group. I configured an inbound rule that allows HTTP traffic on port 80 from any IPv4 address to enter the resources associated with my Nextwork-security-group.

Outbound rules control what traffic my resources are allowed to send out. By default, AWS security groups include an outbound rule that allows all traffic on all ports to and destination (0.0.0/0). This means that my resources can freely send data outward unless I explicitly restrict it. Since I didn't modify the outbound rules, my Nextwork security group keeps this default therefore, all outbound traffic is permitted.



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▶ Details

sg-0a0d2e1421201e9d8 - Nextwork-security-group-public1

Actions ▼

Details

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Nextwork-security-group-public1	sg-0a0d2e1421201e9d8	A Security Group for the Public1 VPC	ypc-0ce89e2934bb6614c
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Network ACLs

Network ACLs (Access Control Lists) are an additional layer of security that apply broad traffic rules at the subnet level. This means that every resource in the subnet is affected. Unlike security groups, Network ACLs support both Allow and Deny rules, making them useful for explicitly blocking traffic from specific IP ranges and ports. They are stateless, meaning inbound and outbound traffic each need their own explicit rules. Allowed inbound traffic does not automatically get a return path out. Rules are evaluated in the order from lowest number, and the first matching rule wins.

Security groups vs. network ACLs

Network ACLs apply broad traffic at the subnet level, affecting every resource within it. They support both Allow and Deny rules and are stateless, meaning that inbound and outbound traffic each need their own explicit rule. Security groups, on the other hand, operate at the individual resources level, offering more control over specific ports and protocols. They only have Allow rules and are stateful, therefore return traffic is automatically permitted without a separate outbound rule. Having both network ACLs and security groups creates a dual layer of security. Traffic must pass through the network ACL at the subnet level before reaching the security group at the resource level, significantly reducing the chances of unwanted access.



Default vs Custom Network ACLs

Similar to security groups, network ACLs use inbound and outbound rules

All default network ACLs have two rules on both the inbound and outbound sides. Rule 100 allows all traffic from any source or destination(0.0.0.0/0), meaning everything is permitted by default. Rule * is a permanent catch-all Deny rule that blocks any traffic that doesn't match a rule above it. This rule can not be edited or deleted.

In contrast, a custom ACL's inbound and outbound rules are automatically set to Deny. Unlike the default NACL which comes with rule 100 allowing all traffic, a custom network ACL only has the Rule * catch-all deny with nothing above it. Traffic remains blocked until you add Allow rules.

acl-Oea6e1605b8213507 / NextWork Network ACL Public1

1

acl-Oea6e1605b8213507 / NextWork Network ACL Public1

Actions

Details
Info

Network ACL ID
 acl-Oea6e1605b8213507

Owner
 479887546653

Associated with
[subnet-0336a7365b50f6139 / my-subnet-public1](#)

Default
No

VPC ID
[vpc-0d028e8649e286e71 / Public1](#)

Inbound rules
Outbound rules
Subnet associations
Tags

Inbound rules (2)

Edit inbound rules

1

Rule number	Type	Protocol	Port range	Source	Allow/Deny
100	All traffic	All	All	0.0.0.0/0	Allow
*	All traffic	All	All	0.0.0.0/0	Deny



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